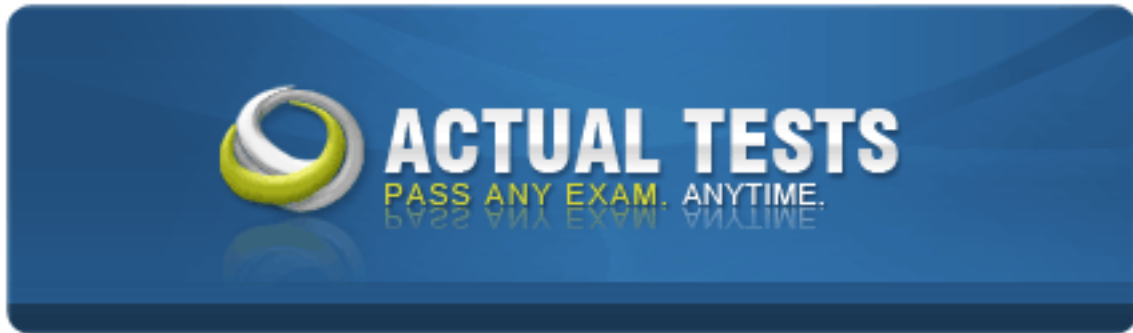


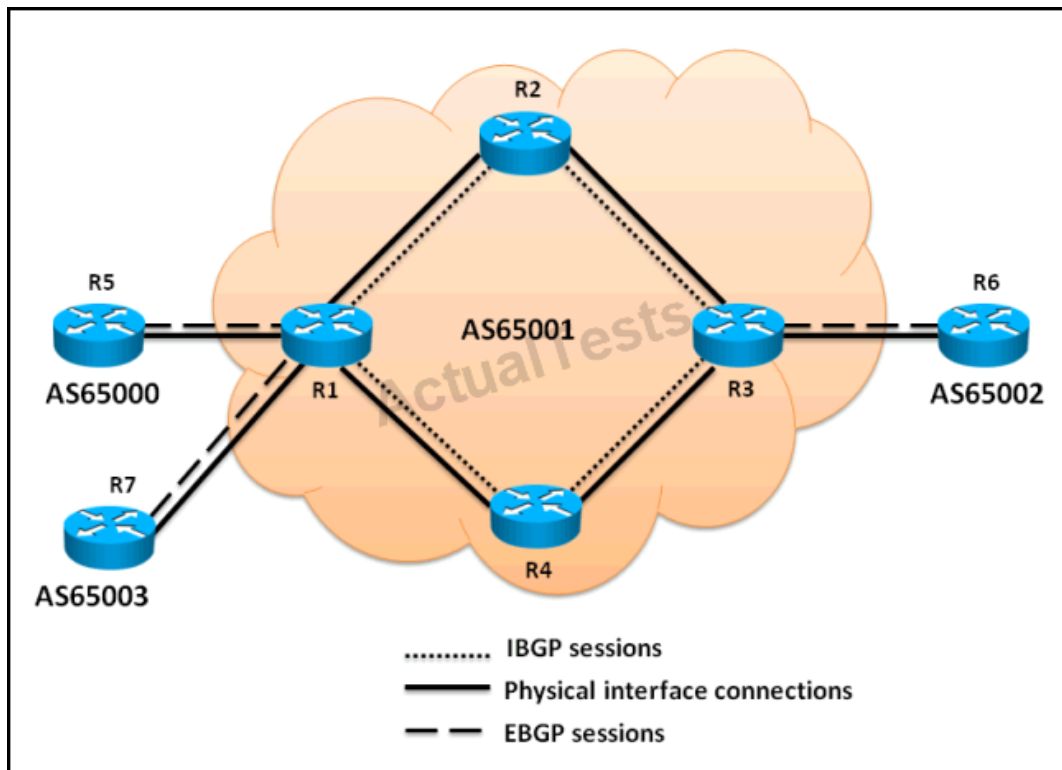
Cisco 642-885



**Deploying Cisco Service Provider Advanced Network
Routing
Version: 4.1**

QUESTION NO: 1

Referring to the topology diagram show in the exhibit,



which three statements are correct regarding the BGP routing updates? (Choose three.)

- A.** The EBGP routing updates received by R1 from R5 will be propagated to the R2, R4, and R7 routers
- B.** The EBGP routing updates received by R3 from R6 will be propagated to the R2 and R4 routers
- C.** The EBGP routing updates received by R1 from R5 will be propagated to the R2 and R4 routers
- D.** The IBGP routing updates received by R3 from R2 will be propagated to the R6 router
- E.** The IBGP routing updates received by R2 from R1 will be propagated to the R3 router
- F.** The IBGP routing updates received by R1 from R4 will be propagated to the R5, R7, and R2 routers

Answer: A,B,D

Explanation:

QUESTION NO: 2

When a BGP route reflector receives an IBGP update from a non-client IBGP peer, the route

reflector will then forward the IBGP updates to which other router(s)?

- A. To the other clients only
- B. To the EBGP peers only
- C. To the EBGP peers and other clients only
- D. To the EBGP peers and other clients and non-clients

Answer: C

Explanation:

QUESTION NO: 3

Which two BGP mechanisms are used to prevent routing loops when using a design with redundant route reflectors? (Choose two.)

- A. Cluster-list
- B. AS-Path
- C. Originator ID
- D. Community
- E. Origin

Answer: A,C

Explanation:

QUESTION NO: 4

Which two statements correctly describe the BGP ttl-security feature? (Choose two.)

- A. This feature protects the BGP processes from CPU utilization-based attacks from EBGP neighbors which can be multiple hops away
- B. This feature prevents IBGP sessions with non-directly connected IBGP neighbors
- C. This feature will cause the EBGP updates from the router to be sent using a TTL of 1
- D. This feature needs to be configured on each participating BGP router
- E. This feature is used together with the ebgp-multihop command

Answer: A,D

Explanation:

QUESTION NO: 5

When implementing source-based remote-triggered black hole filtering, which two configurations are required on the edge routers that are not the signaling router? (Choose two.)

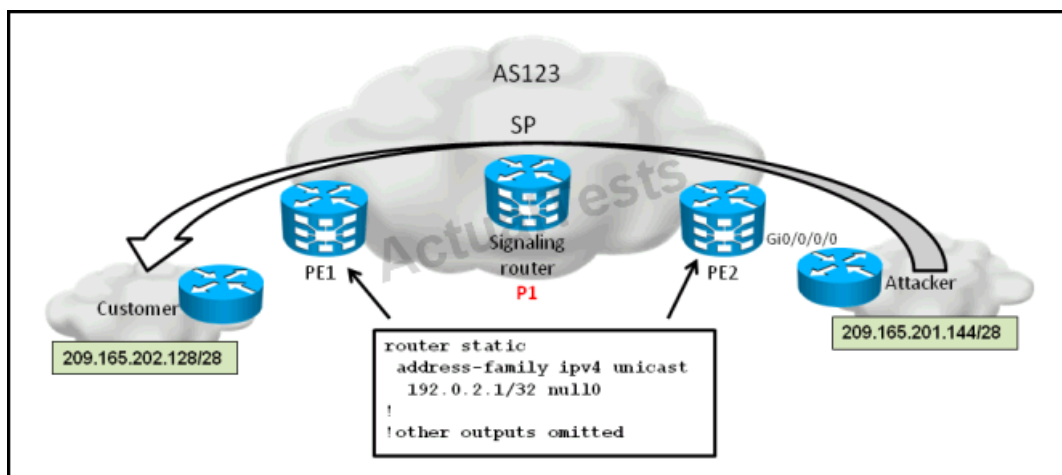
- A. A static route to a prefix that is not used in the network with a next hop set to the Null0 interface
- B. A static route pointing to the IP address of the attacker
- C. uRPF on all external facing interfaces at the edge routers
- D. Redistribution into BGP of the static route that points to the IP address of the attacker
- E. A route policy to set the redistributed static routes with the no-export BGP community

Answer: A,C

Explanation:

QUESTION NO: 6

Refer to the topology diagram shown in the exhibit and the partial configurations shown below.



Once the attack from 209.165.201.144/28 to 209.165.202.128/28 has been detected, which additional configurations are required on the P1 IOS-XR router to implement source-based remote-triggered black hole filtering?

!

router bgp 123

address-family ipv4 unicast

redistribute static route-policy test

!

A. router static

```
address-family ipv4 unicast
209.165.202.128/28 null0 tag 666
192.0.2.1/32 null0 tag 667
```

!

```
route-policy test
if tag is 666 then
set next-hop 192.0.2.1
endif
if tag is 667 then
set community (no-export)
endif
end-policy
```

!

B. router static

```
address-family ipv4 unicast
209.165.201.144/28 null0 tag 666
192.0.2.1/32 null0 tag 667
```

!

```
route-policy test
if tag is 666 then
set next-hop 192.0.2.1
endif
if tag is 667 then
set community (no-export)
endif
end-policy
```

!

C. router static

```
address-family ipv4 unicast
209.165.201.144/28 null0 tag 666
192.0.2.1/32 null0
```

!

```
route-policy test
if tag is 666 then
set next-hop 192.0.2.1
set community (no-export)
endif
end-policy
```

D. router static

```
address-family ipv4 unicast
209.165.202.128/28 null0 tag 666
192.0.2.1/32 null0
```

!

```
route-policy test
if tag is 666 then
set next-hop 192.0.2.1
set community (no-export)
endif
end-policy
!
```

Answer: C

Explanation:

QUESTION NO: 7

In Cisco IOS-XR, the maximum-prefix command, to control the number of prefixes that can be installed from a BGP neighbor, is configured under which configuration mode?

- A. RP/0/RSP0/CPU0:P2(config-bgp)#
- B. RP/0/RSP0/CPU0:P2(config-bgp-af)#
- C. RP/0/RSP0/CPU0:P2(config-bgp-nbr)#
- D. RP/0/RSP0/CPU0:P2(config-bgp-nbr-af)#

Answer: D

Explanation:

QUESTION NO: 8

In Cisco IOS-XR, the ttl-security command is configured under which configuration mode?

- A. RP/0/RSP0/CPU0:P2(config)#
- B. RP/0/RSP0/CPU0:P2(config-bgp)#
- C. RP/0/RSP0/CPU0:P2(config-bgp-nbr)#
- D. RP/0/RSP0/CPU0:P2(config-bgp-af)#
- E. RP/0/RSP0/CPU0:P2(config-bgp-nbr-af)#

Answer: C

Explanation:

QUESTION NO: 9

Refer to the exhibit.

```
router bgp 65123
  bgp cluster-id 17
  address-family ipv4 unicast
  exit
```

Given the partial BGP configuration, which configuration correctly completes the Cisco IOS-XR route reflector configuration where both the 1.1.1.1 and 2.2.2.2 routers are the clients and the 3.3.3.3 router is a non-client IBGP peer?

A. neighbor 1.1.1.1
remote-as 65123
route-reflector-client

neighbor 2.2.2.2
remote-as 65123
route-reflector-client
neighbor 3.3.3.3
remote-as 65123

B. neighbor 1.1.1.1
address-family ipv4 unicast
remote-as 65123
route-reflector-client
neighbor 2.2.2.2
address-family ipv4 unicast
remote-as 65123
route-reflector-client
neighbor 3.3.3.3
address-family ipv4 unicast
remote-as 65123

C. neighbor 1.1.1.1
remote-as 65123
address-family ipv4 unicast
route-reflector-client
neighbor 2.2.2.2
remote-as 65123
address-family ipv4 unicast
route-reflector-client
neighbor 3.3.3.3
remote-as 65123

D. neighbor 1.1.1.1 remote-as 65123
neighbor 1.1.1.1 route-reflector-client
neighbor 2.2.2.2 remote-as 65123

neighbor 2.2.2.2 route-reflector-client
neighbor 3.3.3.3 remote-as 65123

Answer: C

Explanation:

QUESTION NO: 10

Which three methods can be used to reduce the full-mesh IBGP requirement in a service provider core network? (Choose three.)

- A.** implement route reflectors
- B.** enable multi-protocol BGP sessions between all the PE routers
- C.** implement confederations
- D.** implement MPLS (LDP) in the core network on all the PE and P routers
- E.** enable BGP synchronization
- F.** disable the IBGP split-horizon rule

Answer: A,C,D

Explanation:

QUESTION NO: 11

Which type of BGP session behaves like an EBGP session during session establishment but behaves like an IBGP session when propagating routing updates where the local preference, multi-exit discriminator, and next-hop attributes are not changed?

- A.** BGP sessions between a route reflector and its clients
- B.** BGP sessions between a route reflector and its non-client IBGP peers
- C.** BGP sessions between a route reflector and another route reflector
- D.** Intra-confederation IBGP sessions
- E.** Intra-confederation EBGP sessions

Answer: E

Explanation:

QUESTION NO: 12

You noticed a recent change to the BGP configuration on a PE router, the bgp scan time has been

changed from the default value to 30s. Which three effects will this change have? (Choose three.)

- A. The BGP table will be examined and verified more frequently
- B. The BGP keepalive messages will be sent to the BGP peers at a faster rate
- C. The BGP table will be modified more quickly in the event that a next-hop address becomes unreachable
- D. The CPU load of the router will increase
- E. The minimum time interval between sending EBGP and IBGP routing updates will decrease
- F. The BGP convergence time will increase

Answer: A,C,D

Explanation:

QUESTION NO: 13

On Cisco IOS-XR, which BGP process can be distributed into multiple instances?

- A. BGP process manager
- B. BGP RIB process
- C. BGP speaker process
- D. BGP scanner process
- E. BGP dampening process

Answer: C

Explanation:

QUESTION NO: 14

Refer to the configuration exhibit, taken from a Cisco IOS-XR router.

```
!  
router static  
address-family ipv4 unicast  
192.0.2.1/32 Null0  
!  
route-policy RTBH  
if tag is 666 then  
set next-hop 192.0.2.1  
endif  
end-policy  
!  
router bgp 65123  
address-family ipv4 unicast  
redistribute static route-policy RTBH  
!  
!When attacks are detected from 209.165.201.144/28  
!  
router static  
address-family ipv4 unicast  
209.165.201.144/28 null0 tag 666  
!
```

Which configuration change is required to properly enable this router as the signaling router for implementing source-based RTBH filtering?

- A. set community (no-export) in the route policy
- B. pass in the route policy
- C. set local-preference 1000 in the route policy
- D. The 192.0.2.1/32 static route should be tagged as 666 (tag 666)

Answer: A

Explanation:

QUESTION NO: 15

Refer to the Cisco IOS-XR BGP configuration exhibit.

```
!  
route-policy passall  
permit  
end-policy  
!  
router bgp 65123  
af-group abc address-family ipv4 unicast  
route-policy passall in  
route-policy passall out  
!  
neighbor-group efg  
password C!sc0!3o  
ttl-security  
update-source Loopback0  
maximum-prefix 10  
address-family ipv4 unicast  
use af-group abc  
!  
neighbor 209.165.201.130  
remote-as 65234  
use neighbor-group efg  
!
```

Identify two configuration errors. (Choose two.)

- A. The neighbor-group efg is missing the ebgp-multihop 2 configuration
- B. The ttl-security configuration command is missing the option to set the number of hops
- C. The passall route policy is wrong
- D. The route-policy passall in and route-policy passall out commands should be configured under the neighbor-group efg instead of the af-group abc
- E. The maximum-prefix 10 configuration should be configured under the af-group abc instead of the neighbor-group efg

Answer: C,E

Explanation:

QUESTION NO: 16

Refer to the Cisco IOS-XR show output exhibit.

```
RP/0/RSP0/CPU0:P1#show bgp neighbors 10.1.1.1 configuration
Wed Oct 26 17:45:09.690 UTC
neighbor 10.1.1.1
  remote-as 64500          [ ]
  update-source Loopback0 [ ]
  address-family IPv4 Unicast [ ]
```

Which statement is correct?

- A. The [] indicates the configuration has a problem
- B. The [] indicates the 10.1.1.1 neighbor peering session has not been established
- C. The [] indicates the configuration was not inherited from a group
- D. The [] indicates the configuration has not been committed
- E. The [] indicates the corresponding BGP peer configuration has a mismatch configuration

Answer: C

Explanation:

QUESTION NO: 17

Which two statements correctly describe the RPF check when a multicast packet arrives at a router? (Choose two.)

- A. The router looks up the source address in the unicast routing table to determine if the packet has arrived on the interface that is on the reverse path back to the source
- B. The router looks up the destination address in the unicast routing table to determine if the packet has arrived on the interface that is on the reverse path back to the destination
- C. If the packet has arrived on the interface leading back to the destination, the RPF check passes and the packet is forwarded. If the RPF check fails, the packet is dropped
- D. If the packet has arrived on the interface leading back to the source, the RPF check passes and the packet is forwarded. If the RPF check fails, the packet is dropped

Answer: A,D

Explanation:

QUESTION NO: 18

When enabling interdomain multicast routing, which two statements are correct? (Choose two.)

- A. Multiprotocol BGP is used instead of PIM SM to build the intradomain and interdomain multicast distribution trees
- B. Use MSDP to enable the RPs from different domains to exchange information about active multicast sources
- C. MSDP SA packets are sent between the multiprotocol BGP peers
- D. Noncongruent unicast and multicast topologies can be supported using multiprotocol BGP

Answer: B,D

Explanation:

QUESTION NO: 19

The 224.192.16.1 multicast IP address maps to which multicast MAC address?

- A. 01-00-5E-C0-10-01
- B. 01-00-5E-40-10-01
- C. 01-00-5E-00-10-01
- D. 01-00-5E-C0-16-01

Answer: B

Explanation:

QUESTION NO: 20

The following Cisco IOS-XR configuration command will globally enable which multicast process(es) on the router?

RP/0/RP0/CPU0:router(config)# multicast-routing

- A. IGMP only
- B. PIM only
- C. IGMP and MLD only
- D. PIM and IGMP only
- E. PIM and IGMP and MLD

Answer: E

Explanation:

QUESTION NO: 21

Which three statements are correct regarding PIM-SM? (Choose three.)

- A. There are three ways to configure the RP: Static RP, Auto-RP, or BSR
- B. PIM-SM only uses the RP rooted shared tree and has no option to switch over to the shortest path tree
- C. Different RPs can be configured for different multicast groups to increase RP scalability
- D. Candidate RPs and RP mapping agents are configured to enable Auto-RP
- E. PIM-SM uses the implicit join model

Answer: A,C,D

Explanation:

QUESTION NO: 22

Which of the following is a feature added in IGMPv3?

- A. Support for source filtering
- B. Support for Host Membership Report and a Leave Group message
- C. Uses a new variation of the Host Membership Query called the Group-Specific Host Membership Query
- D. Uses an election process to determine the querying router on the LAN
- E. Uses an election process to determine the designated router on the LAN
- F. IPv6 support

Answer: A

Explanation:

QUESTION NO: 23

Which types of multicast distribution tree can PIM-SM use?

- A. Only shared tree rooted at the source
- B. Only shared tree rooted at the RP
- C. Only shortest path tree rooted at the RP
- D. Shared tree rooted at the source and shortest path tree switchover
- E. Shared tree rooted at the RP and shortest path tree switchover

F. Shared tree rooted at the first-hop router and shortest path tree rooted at the RP

Answer: E

Explanation:

QUESTION NO: 24

Which multicast routing protocol is most optimal for supporting many-to-many multicast applications?

- A. PIM-SM
- B. PIM-BIDIR
- C. MP-BGP
- D. DVMRP
- E. MSDP

Answer: B

Explanation:

QUESTION NO: 25

Which statement is correct regarding using the TTL threshold to define the delivery boundaries of multicast traffic?

- A. If a packet TTL is less than the specified TTL threshold, the packet is forwarded out of the interface
- B. If a packet TTL is greater or equal to the specified TTL threshold, the packet is forwarded out of the interface
- C. If a packet TTL is equal to the specified TTL threshold, the packet is dropped
- D. When a multicast packet arrives, the TTL threshold value is decremented by 1. If the resulting TTL threshold value is greater than or equal to 0, the packet is dropped

Answer: B

Explanation:

QUESTION NO: 26

Refer to the exhibit.

```
interface loopback 0
  ipv4 address 10.0.0.1/24
  no shutdown
!
interface loopback 1
  ipv4 address 10.2.0.1/24
  no shutdown
!
ipv4 access-list acl1
  10 permit 224.11.11.11 0.0.0.0 any
!
ipv4 access-list acl2
  10 permit 224.99.99.99 0.0.0.0 any
!
multicast-routing
  interface all enable
!
router pim
  auto-rp mapping-agent loopback 0 scope 15 interval 60
  auto-rp candidate-rp loopback 0 scope 15 group-list acl1 interval 60 bidir
  auto-rp candidate-rp loopback 1 scope 15 group-list acl2 interval 60
!
end
```

Which three statements are correct regarding the Cisco IOS-XR configuration? (Choose three.)

- A. This router, acting as the RP mapping agent, will send RP announcement messages to the 224.0.1.40 group
- B. This router, acting as the RP mapping agent, will send RP discovery messages to the 224.0.1.39 group
- C. This router is the RP mapping agent only for the 224.11.11.11 and 224.99.99.99 multicast groups
- D. This router is a candidate PIM-SM RP for the 224.99.99.99 multicast group
- E. This router is a candidate PIM-BIDIR RP for the 224.11.11.11 multicast group
- F. IGMPv3 is enabled on all interfaces
- G. Other routers will recognize this router as the RP for all multicast groups with this router loopback 0 IP address

Answer: D,E,F

Explanation:

QUESTION NO: 27

Which statement is correct regarding MP-BGP?

- A. MP-BGP can indicate whether an advertised prefix (NLRI) is to be used for unicast routing,

multicast RPF checks or for both using different SAFIs.

B. MP-BGP uses a single BGP table to maintain all the unicast prefixes for unicast forwarding and all the unicast prefixes for RPF checks.

C. MP-BGP can be used to propagate multicast state information, which eliminates the need to use PIM for building the multicast distribution trees.

D. MP-BGP enables BGP to carry IP multicast routes used by MSDP to build the multicast distribution trees.

Answer: A

Explanation:

QUESTION NO: 28

When implementing interdomain multicast routing, which mechanism can be used to advertise multicast sources in one domain to the other domains, allowing the RPs to build interdomain multicast distribution trees?

A. Multiprotocol BGP

B. PIM

C. MSDP

D. Auto RP

E. BSR

F. MLD

Answer: C

Explanation:

QUESTION NO: 29

Which two statements regarding Auto RP operations and implementations are correct? (Choose two.)

A. Candidate RPs send RP announcements to the 224.0.1.39 multicast group, and the mapping agents send RP discovery messages to the 224.0.1.40 multicast group

B. Every PIM-SM router must be configured with the RP mapping agent IP address

C. Candidate RPs learn the IP address of the mapping agents via periodic RP discovery messages

D. Administrative scoping can be configured to limit the scope of the RP announcements

E. A Reverse Path Forwarding check is done on the RP discovery messages

F. RP discovery messages are flooded hop by hop throughout the network as multicast to the all PIM routers multicast group with a TTL of 1

Answer: A,D

Explanation:

QUESTION NO: 30

What is determined by running the same hash algorithm on all PIMv2 routers?

- A. The SPT from the RP to the multicast source
- B. The SPT from the last hop router to the multicast source
- C. Auto RP election
- D. Which BSR to use for a particular multicast group
- E. Which RP to use from a set of candidate RPs in the RP set

Answer: E

Explanation:

QUESTION NO: 31

When configuring PIM operations, what is the effect of setting the SPT threshold to infinity?

- A. The multicast source to the RP path will never switch over to the shortest path tree
- B. All the PIM routers will have more (S,G) states, thus consuming more router resources
- C. The receivers will be able to immediately switch over to the shortest path tree after receiving the first multicast packets on the shared tree via the RP
- D. The last-hop routers will never switch over to the shortest path tree and will always remain on the shared tree

Answer: D

Explanation:

QUESTION NO: 32

Which multicast group range is reserved for SSM?

- A. 224.0.0.0/8
- B. 225.0.0.0/8
- C. 232.0.0.0/8
- D. 239.0.0.0/8

Answer: C

Explanation:

QUESTION NO: 33

Which four statements are correct regarding MSDP configurations and operations? (Choose four.)

- A. The MSDP peers are also typically the RPs in respective routing domains.
- B. SA messages are flooded to all other MSDP peers without any restrictions
- C. On Cisco IOS, IOS-XE, and IOS-XR, the router can be configured to cache the SA messages to reduce the join latency
- D. SA messages are used to advertise active sources in a domain
- E. MSDP establishes neighbor relationships with other MSDP peers using TCP port 639
- F. MSDP peerings on Cisco IOS, IOS-XE, and IOS-XR support MD5 or SHA1 authentication

Answer: A,C,D,E

Explanation:

QUESTION NO: 34

Assume that the R1 router is enabled for PIM-SM and receives a multicast packet sourced from 172.16.1.100, and the R1 router has multicast receivers on the Gi0/1, Gi0/2, Gi0/3 and Gi0/4 interfaces.

R1 routing table:	
172.16.1.0/24	via Gi0/1
172.16.2.0/24	via Gi0/2
172.16.3.0/24	via Gi0/3
0.0.0.0/0	via Gi0/4

The multicast packet from the 172.16.1.100 source must arrive on which interface on the R1 router for it to be forwarded out the other interfaces?

- A. Gi0/1
- B. Gi0/2
- C. Gi0/3
- D. Gi0/4

- E. Gi0/1 or Gi0/2 or Gi0/3 or Gi0/4
- F. Gi0/2 or Gi0/3
- G. Gi0/1 or Gi0/4

Answer: A

Explanation:

QUESTION NO: 35

When verifying multicast configurations and operations on Cisco IOS-XR routers, which two statements regarding show commands are correct? (Choose two.)

- A. Use the show route ipv4 multicast command to display the incoming and outgoing interface lists for each of the joined multicast groups
- B. Use the show pim rpf command to display the RPF information for the RP or for the multicast source
- C. Use the show mrib route command to display the (*,G) and (S,G) states information on the router
- D. Use the show mrib route command to display the configured static multicast routes

Answer: B,C

Explanation:

QUESTION NO: 36

The bsr-border router PIM interface configuration command is used for what purpose?

- A. To enable the router as the candidate RP
- B. To enable the router as the candidate BSR
- C. To enable the router as the BSR mapping agent
- D. To set up an administrative boundary to prevent BSR messages from being sent out through an interface
- E. To define a boundary to restrict the RP discovery and announcement messages from being sent outside the PIM-SM domain

Answer: D

Explanation:

QUESTION NO: 37

When implementing Anycast RP, the RPs are also required to establish which kind of peering with each other?

- A. BGP
- B. Multiprotocol BGP
- C. MSDP
- D. Bidirectional PIM
- E. PIM SSM

Answer: C

Explanation:

QUESTION NO: 38

Refer to the Cisco IOS-XR show output exhibit.

```
RP/0/RSP0/CPU0:PE1#show mrib route
Thu Dec 1 19:14:38.044 UTC IP Multicast Routing Information Base
Entry flags: L - Domain-Local Source, E - External Source to the Domain,
             C - Directly-Connected Check, S - Signal, IA - Inherit Accept,
             IF - Inherit From, D - Drop, MA - MDT Address, ME - MDT Encap,
             MD - MDT Decap, MT - MDT Threshold Crossed, MH - MDT interface handle
             CD - Conditional Decap, MPLS - MPLS Decap, MF - MPLS Encap, EX - Extranet
             MoFE - MoFRR Enabled, MoFS - MoFRR State
Interface flags: F - Forward, A - Accept, IC - Internal Copy,
                NS - Negate Signal, DP - Don't Preserve, SP - Signal Present,
                II - Internal Interest, ID - Internal Disinterest, LI - Local Interest,
                LD - Local Disinterest, DI - Decapsulation Interface
                EI - Encapsulation Interface, MI - MDT Interface, LVIF - MPLS Encap,
                EX - Extranet, A2 - Secondary Accept

<output omitted>

(*,224.1.1.1) RPF nbr: 192.168.11.1 Flags: C
Up: 14:34:53
Incoming Interface List
  GigabitEthernet0/0/0/2 Flags: A NS, Up: 14:34:53
Outgoing Interface List
  Loopback0 Flags: F IC NS II LI, Up: 14:34:53
  GigabitEthernet0/0/0/0 Flags: F NS, Up: 14:34:33
```

Which two statements are correct? (Choose two.)

- A. The RPF neighbor 192.168.11.1 is the path towards the RP for the 224.1.1.1 multicast group

- B. The RP for the 224.1.1.1 multicast group is reachable over the Gi0/0/0/0 interface
- C. This router is the RP for the 224.1.1.1 multicast group
- D. Incoming 224.1.1.1 multicast group traffic will be sent out through the Gi0/0/0/0 interface
- E. Incoming 224.1.1.1 multicast group traffic will be sent out through the Gi0/0/0/2 interface

Answer: A,D

Explanation:

QUESTION NO: 39

What must occur before an (S,G) entry can be populated in the multicast routing table?

- A. The (*,G) entry must have timed out
- B. The (*,G) entry OIL must be null
- C. The router must be directly connected to the multicast source
- D. The parent (*,G) entry must be created first

Answer: D

Explanation:

QUESTION NO: 40

Which field in the IPv6 header can be used to set the DSCP value?

- A. Flow Label
- B. Type of Service
- C. Traffic Class
- D. Precedence
- E. EXP

Answer: C

Explanation:

QUESTION NO: 41

Which mechanism is used by an IPv6 multicast receiver to join an IPv6 multicast group?

- A. IGMP report

- B. IGMP join
- C. MLD report
- D. General query
- E. PIM join

Answer: C

Explanation:

QUESTION NO: 42

Which of the following can be used by dual-stack service providers supporting IPv4/IPv6 customers with dual-stack hosts using public IPv6 addresses and private IPv4 addresses?

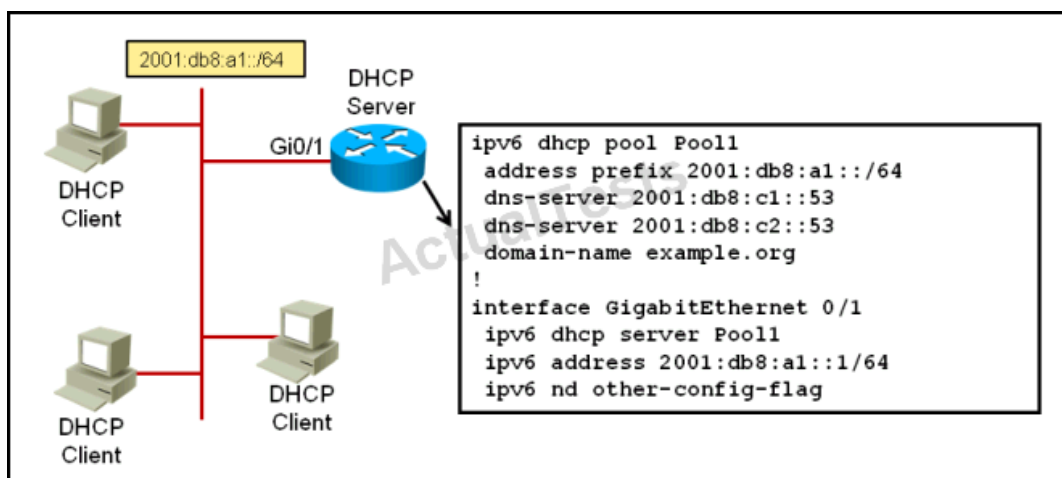
- A. NAT64
- B. 6RD
- C. 6to4 tunnels
- D. Carrier-grade NAT

Answer: D

Explanation:

QUESTION NO: 43

Refer to the Cisco IOS DHCPv6 configuration shown in the exhibit.



Which statement is correct?

- A.** The configuration is missing a command under interface Gi0/1 to indicate to the attached hosts to use stateful DHCPv6 to obtain their IPv6 addresses
- B.** The IPv6 router advertisements indicate to the attached hosts on the Gi0/1 interface to get other information besides their IPv6 address via stateless auto configuration
- C.** The IPv6 DHCPv6 server pool configuration is misconfigured
- D.** The DNS server address can also be imported from another upstream DHCPv6 server

Answer: A

Explanation:

QUESTION NO: 44

Which IPv6 mechanism occurs between a provider edge router and the customer premises equipment router to allow an ISP to automate the process of assigning a block of IPv6 addresses to a customer for use within the customer network?

- A.** Router Advertisement
- B.** DHCPv6 Prefix Delegation
- C.** DHCPv6 Lite
- D.** Stateful DHCPv6

Answer: B

Explanation:

QUESTION NO: 45

Which three statements regarding NAT64 operations are correct? (Choose three.)

- A.** With stateful NAT64, many IPv6 address can be translated into one IPv4 address, thus IPv4 address conservation is achieved
- B.** Stateful NAT64 requires the use of static translation slots so IPv6 hosts and initiate connections to IPv4 hosts.
- C.** With stateless NAT64, the source and destination IPv4 addresses are embedded in the IPv6 addresses
- D.** NAT64 works in conjunction with DNS64
- E.** Both the stateful and stateless NAT64 methods will conserve IPv4 address usage

Answer: A,C,D

Explanation:

QUESTION NO: 46

Which type of DNS record is used for IPv6 forward lookups?

- A. A records
- B. AAAA records
- C. PTR records
- D. MX records

Answer: B

Explanation:

QUESTION NO: 47

What is enabled by default on Cisco IOS-XR routers and cannot be disabled?

- A. SSH server
- B. Multicast routing
- C. IPv4 and IPv6 CEF
- D. IPv6 routing
- E. CDP
- F. BFD

Answer: C

Explanation:

QUESTION NO: 48

The IPv6 2002::/16 prefix is used in which kind of implementations?

- A. 6RD
- B. 6to4
- C. NAT64
- D. IPv6 Multicast

Answer: B

Explanation:

QUESTION NO: 49

When implementing IP SLA icmp-echo probes on Cisco IOS-XE routers, which two options are available for IPv6? (Choose two.)

- A. flow-label
- B. hop-limit
- C. DSCP
- D. traffic-class
- E. TOS

Answer: A,D

Explanation:

QUESTION NO: 50

With IPv6 multicast, which feature can be used as a replacement method for static RP configuration?

- A. PIM Snooping
- B. MLD
- C. MLD Snooping
- D. Embedded RP
- E. DHCPv6

Answer: D

Explanation:

QUESTION NO: 51

Which additional feature is provided using MLDv2 that is not available in MLDv1?

- A. Multicast Address Specific Queries
- B. Source filtering
- C. Done messages
- D. Report messages

Answer: B

Explanation:

QUESTION NO: 52

When implementing high-availability stateful switchover BGP routing, in which situation would Cisco NSR be required?

- A.** On the PE routers connecting to the CE routers which are not NSF aware or are not NSF capable
- B.** On the PE routers connecting to the CE routers which support graceful restart
- C.** On the PE routers connecting to the CE routers which are incapable of performing stateful switchover operations because the CE routers are only NSF aware but not NSF capable
- D.** On the PE routers connecting to the CE routers which are incapable of performing stateful switchover operations because the CE routers are only NSF capable but not NSF aware
- E.** On the service provider core P routers which are also NSF aware
- F.** On the service provider core P routers which are also NSF capable

Answer: A

Explanation:

QUESTION NO: 53

Refer to the exhibit.



```
router bgp 65123
bgp graceful-restart
```

Which statement correctly explains the bgp graceful-restart command?

- A.** This command is used to enable NSR and is entered on the NSR-capable router, and also on any NSR-aware peer
- B.** This command is used to enable NSF and is entered on the NSF-capable router, and also on any NSF-aware peer
- C.** This command is only required on the NSF-capable routers to enable BGP graceful restart with the BGP peers
- D.** This command is only required on the NSF-aware routers to enable BGP graceful restart with the BGP peers
- E.** This command is only required on the NSR-capable routers to enable BGP graceful restart with the BGP peers

Answer: B

Explanation:

QUESTION NO: 54

Refer to the exhibit.

```
router bgp 64500
bfd multiplier 2
bfd minimum-interval 20
address-family ipv4 unicast
network 10.1.1.0/24
!
address-family ipv6 unicast
!
neighbor 192.168.1.1
remote-as 65001
address-family ipv4 unicast
!
end
```

Which configuration is missing to complete the configuration task of enabling BFD with the 192.168.1.1 EBGp peer?

- A.** bfd fast-detect also needs to be enabled globally under router bgp 64500
RP/0/RSP0/CPU0:P1(config-bgp)#bfd fast-detect
- B.** bfd fast-detect also needs to be enabled for the address-family under address-family ipv4 unicast
RP/0/RSP0/CPU0:P1(config-bgp-af)#bfd fast-detect
- C.** bfd fast-detect also needs to be enabled for the 192.168.1.1 neighbor under neighbor 192.168.1.1
RP/0/RSP0/CPU0:P1(config-bgp-nbr)#bfd fast-detect
- D.** bfd fast-detect also needs to be enabled for the 192.168.1.1 neighbor address-family under neighbor 192.168.1.1 address-family ipv4 unicast
RP/0/RSP0/CPU0:P1(config-bgp-nbr-af)#bfd fast-detect
- E.** bfd fast-detect also needs to be enabled globally on the router

RP/0/RSP0/CPU0:P1(config)#bfd fast-detect

Answer: C

Explanation:

QUESTION NO: 55

Which two features are used to provide high availability multicast? (Choose two.)

- A. BFD
- B. NSF/SSO
- C. PIM NSR
- D. PIM triggered join
- E. IGMP triggered report
- F. MSDP

Answer: B,D

Explanation:

QUESTION NO: 56

What are three BGP configuration characteristics of a multihomed customer that is connected to multiple service providers? (Choose three.)

- A. The multihomed customer can use local preference to influence the return traffic from the service providers
- B. The multihomed customer announces its assigned IP address space to its service providers through BGP
- C. The multihomed customer has to decide whether to perform load sharing or use a primary/backup implementation
- D. The multihomed customer must use private AS number
- E. The multihomed customer configures outbound route filters to prevent itself from becoming a transit AS

Answer: B,C,E

Explanation:

QUESTION NO: 57

Refer to the EBGP configuration on a PE IOS-XR router exhibit.

After the EBGp configuration, no routes are accepted from the EBGp peer, nor are any routes advertised to the EBGp peer.

```
router bgp 65001
 address-family ipv4 unicast
  network 172.16.1.0/24
  network 192.168.1.0/24
 !
 neighbor 10.1.1.1
  remote-as 65002
 !
```

What could be the problem?

- A. The update-source neighbor configuration command must also be configured
- B. The next-hop-self neighbor configuration command must also be configured
- C. EBGp neighbors must have an inbound and outbound route policy configured
- D. An access list is blocking IP protocol 179 packets between the two EBGp peers
- E. The maximum-prefix neighbor configuration command must also be configured

Answer: C

Explanation:

QUESTION NO: 58

Refer to the exhibit.

224.10.0.1
224.138.0.1
225.10.0.1
225.138.0.1
226.10.0.1
226.138.0.1
227.10.0.1
227.138.0.1
228.10.0.1
228.138.0.1
229.10.0.1
229.138.0.1
230.10.0.1
230.138.0.1
231.10.0.1
231.138.0.1
232.10.0.1
232.138.0.1
233.10.0.1
233.138.0.1
234.10.0.1
234.138.0.1
235.10.0.1
235.138.0.1
236.10.0.1
236.138.0.1
237.10.0.1
237.138.0.1
238.10.0.1
238.138.0.1
239.10.0.1
239.138.0.1

The following multicast IP addresses map to which multicast MAC address?

- A. 01:00:5E:8A:00:01
- B. 01:00:5E:0A:00:01
- C. 01:00:5E:7A:00:01
- D. 01:00:5E:05:00:01

Answer: B

Explanation:

QUESTION NO: 59

A junior network engineer has just configured a new IBGP peering between two Cisco ASR9K PE routers in the network using the loopback interface of the router, but the IBGP neighborhood is not able to be established. Which two verification steps will be helpful in troubleshooting this problem? (Choose two.)

- A.** Verify that the network command under router BGP is configured correct on each router for announcing the router's loopback interface in BGP
- B.** Verify that the ibgp-multihop command under the BGP neighbor is configured correctly on each router
- C.** Verify that the loopback interfaces are reachable over the IGP
- D.** Verify that the update-source loopback command under the BGP neighbor is configured correctly on each router
- E.** Verify that the ttl-security command under the BGP neighbor is configured correctly on each router to enable the router to send the BGP packets using a proper TTL value
- F.** Verify that the UDP port 179 traffic is not being blocked by an ACL or firewall between the two IBGP peers

Answer: C,D

Explanation:

QUESTION NO: 60

Refer to the exhibit for the outputs from an ASR9K router.


```
RP/0/RSP0/CPU0:PE1#show route ipv6
```

```
Wed Oct 26 20:57:46.433 UTC
```

```
Codes: C - connected, S - static, R - RIP, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
i - ISIS, L1 - IS-IS level-1, L2 - IS-IS level-2
```

```
ia - IS-IS inter area, su - IS-IS summary null, * - candidate default
```

```
U - per-user static route, o - ODR, L - local, G - DAGR
```

```
A - access/subscriber, (I) - FRR Backup path
```

```
Gateway of last resort is not set
```

```
L 2001::db8:10:1:1::1/128 is directly connected,  
09:20:18, Loopback0
```

```
i L2 2001::db8:10:1:10::1/128
```

```
[105/20] via fe80::eab7:48ff:fe2c:a180, 07:59:22, GigabitEthernet0/0/0/0
```

```
C 2001::db8:192:168:101::/80 is directly connected,  
1d05h, GigabitEthernet0/0/0/0
```

```
L 2001::db8:192:168:101::10/128 is directly connected,  
1d05h, GigabitEthernet0/0/0/0
```

```
RP/0/RSP0/CPU0:PE1#ping 2001::db8:10:1:10::1/128
```

```
Wed Oct 26 20:58:01.969 UTC
```

```
%Bad hostname or protocol not running
```

Why did the ping fail?

- A. The ping command is missing the ipv6 option: ping ipv6 2001::db8:10:1:10::1/128
- B. There is a problem with the IS-IS configurations
- C. The fe80::eab7:48ff:fe2c:a180 next-hop is not reachable
- D. The prefix length should be removed from the IPv6 address in the ping command:
ping ipv6 2001::db8:10:1:10::1
- E. IPv6 is not enabled on the Gi0/0/0/0 interface
- F. The IPv6 neighbor discovery protocol is not enabled on the Gi0/0/0/0 interface

Answer: D

Explanation:

QUESTION NO: 61

Which multicast routing protocol supports dense mode, sparse mode and bidirectional mode?

- A. DVMRP

- B. MOSPF
- C. PIM
- D. MP-BGP
- E. MSDP

Answer: C

Explanation:

QUESTION NO: 62

When configuring BFD, the multiplier configuration option is used to determine which value?

- A. The retry interval
- B. The number of BFD packets that can be lost before the BFD peer is declared "down"
- C. The minimum interval between packets accepted from the BFD peers
- D. The number of BFD echo packets that will be originated by the router
- E. The number of routing protocols that will use BFD for fast peer failure detection

Answer: B

Explanation:

QUESTION NO: 63

After configuring the tunnel interface as shown in the exhibit, no IPv6 traffic is passed over the IPv4 network.

```
interface Tunnel0
ipv6 address 2001:db8:3::1/64
tunnel source GigabitEthernet0/0
tunnel destination 209.165.201.6
tunnel mode ipv6ip
```

Which additional configuration is required to pass the IPv6 traffic over the IPv4 network?

- A. Configure an IPv4 address on the tunnel0 interface
- B. Configure an IPv6 static route to send the required IPv6 traffic over the tunnel0 interface
- C. The tunnel destination should be pointing to an IPv6 address instead of an IPv4 address
- D. The tunnel0 interface IPv6 address must use the 2002::/16 prefix

Answer: B

Explanation:

QUESTION NO: 64

Refer to the Cisco IOS configuration exhibit.

```
interface Gi0/0
 ip multicast boundary 1
 !
 access-list 1 deny 224.0.1.39
 access-list 1 deny 224.0.1.40
```

Which statement is correct?

- A.** This configuration is typically configured on the boundary routers within a PIM SM domain to filter out malicious candidate-RP-announce and candidate-RP-discovery packets
- B.** This configuration is typically configured on the RPs within a PIM-SM domain to restrict the candidate-RP-announce packets
- C.** This configuration is typically configured on the mapping agents within a PIM-SM domain to restrict the candidate-RP-discovery packets
- D.** This configuration is typically configured on the MSDP peering routers within a PIM-SM domain to filter out malicious MSDP SA packets

Answer: A

Explanation:

QUESTION NO: 65

Given the IPv6 address of 2001:0DB8::1:800:200E:88AA, what will be its corresponding the solicited-node multicast address?

- A.** FF01::1:200E:88AA
- B.** FF01::1:FF0E:88AA
- C.** FF01:0DB8::1:800:200E:88AA
- D.** FF02::1:FF0E:88AA
- E.** FF02::1:200E:88AA
- F.** FF02:0DB8::1:800:200E:88AA

Answer: D

Explanation:

QUESTION NO: 66

With PIM-SM operations, which four pieces of information are maintained in the multicast routing table for each (*,G) or (S,G) entry? (Choose four.)

- A. RPF Neighbor
- B. RP Set
- C. Incoming Interface
- D. OIL
- E. DF priority
- F. PIM SM state flags

Answer: A,C,D,F

Explanation:

QUESTION NO: 67

What is one of the configuration errors within an AS that can stop a Cisco IOS-XR router from announcing certain prefixes to its EBGp peers?

- A. Some prefixes were mistagged with the no-export BGP community
- B. Some prefixes were set with an MED of 0
- C. The outbound BGP route policy only has set actions defined without any pass actions defined
- D. The inbound BGP route policy only has set actions defined without any pass actions defined

Answer: A

Explanation:

QUESTION NO: 68

Refer to the Cisco IOS-XR configuration exhibit.

```
multicast-routing
!
interface Loopback0
  ipv4 address 10.3.1.1 255.255.255.255
!
interface GigabitEthernet0/0/0/0
  ipv4 address 192.168.103.30 255.255.255.0
  no shut
!
interface GigabitEthernet0/0/0/1
  ipv4 address 192.168.156.50 255.255.255.0
  no shut
!
router isis 1
  net 49.0005.0100.0300.1001.00
  address-family ipv4 unicast
  !
  interface Loopback0
    address-family ipv4 unicast
  !
  interface GigabitEthernet0/0/0/0
    address-family ipv4 unicast
  !
  interface GigabitEthernet0/0/0/1
    address-family ipv4 unicast
  !
router pim
  address-family ipv4
  auto-rp mapping-agent Loopback0 scope 16
  auto-rp candidate-rp Loopback0 scope 16
  !
  interface Loopback0
    enable
  interface GigabitEthernet0/0/0/0
    enable
  interface GigabitEthernet0/0/0/1
    enable
  !
```

The Cisco IOS-XR router is unable to establish any PIM neighbor relationships. What is wrong with the configuration?

- A. The configuration is missing:
interface gi0/0/0/0

```
ip pim sparse-mode
interface gi0/0/0/1
ip pim sparse-mode
interface loopback0
ip pim sparse-mode
```

B. The configuration is missing:

```
multicast-routing
address-family ipv4
interface gi0/0/0/0
enable
interface gi0/0/0/1
enable
```

C. The auto-rp scoping configurations should be set to 1 not 16

D. The RP address has not been configured using the rp-address router PIM configuration command

E. PIM defaults to dense mode operations only, so PIM sparse mode must be enabled using the pim sparse-mode router PIM configuration command

Answer: B

Explanation:

QUESTION NO: 69

On Cisco IOS-XR, which BGP configuration group allows you to define address-family independent commands and address-family dependent commands for each address family?

- A.** neighbor-group
- B.** session-group
- C.** af-group
- D.** peer-group

Answer: A

Explanation:

QUESTION NO: 70 DRAG DROP

Drag the IPv6 tunneling mechanisms on the left to match the correct manual or automatic tunneling category on the right.

IPv6-in-IPv4	Manually configured tunnel	Target
6to4		Target
6RD		
GRE		
	Automatic tunnel	Target
		Target

Answer:

Drag the IPv6 tunneling mechanisms on the left to match the correct manual or automatic tunneling category on the right.

IPv6-in-IPv4	Manually configured tunnel	6RD
6to4		GRE
6RD	Automatic tunnel	6to4
GRE		IPv6-in-IPv4

Explanation: Manually configured tunnel - 6RD , GRE

Automatic Tunnel - 6 to 4 , IPV6-in-IPV4

QUESTION NO: 71 DRAG DROP

Drag the IP multicast characteristic on the left to match the correct multicast service model on the right.

Supports SPT switchover	Any Source Multicast (ASM) service model	Target
Requires IGMPv3 support		Target
Uses (*,G) joins as well as (S,G) joins		Target
No shared trees Only (S,G) state is built between the source and the receiver	Source Specific Multicast (SSM) service model	Target
Uses RPs as the root of the shared tree for a multicast group		Target
Hosts learn the multicast source address via out-of-band mechanism		Target

Answer:

Drag the IP multicast characteristic on the left to match the correct multicast service model on the right

Supports SPT switchover	Any Source Multicast (ASM) service model
Requires IGMPv3 support	Uses RPs as the root of the shared tree for a multicast group
Uses (*,G) joins as well as (S,G) joins	No shared trees Only (S,G) state is built between the source and the receiver
No shared trees Only (S,G) state is built between the source and the receiver	Supports SPT switchover
Uses RPs as the root of the shared tree for a multicast group	Source Specific Multicast (SSM) service model
Hosts learn the multicast source address via out-of-band mechanism	Uses (*,G) joins as well as (S,G) joins
	Requires IGMPv3 support
	Hosts learn the multicast source address via out-of-band mechanism

Explanation:

Any Source Multicast - Uses RP's as the root of the shared tree for a multicast group, ONLY (S,G) state is built between the source and the receiver, Supports SPT Switchover

Source Specific Multicast - Uses (*,G) joins as well as (S,G) Joins, Requires IGMPV3 Support, Hosts learn the multicast source address via out-of-band mechanism

QUESTION NO: 72 DRAG DROP

Referring to the **bgp dampening** command shown below.

Drag the BGP route dampening configuration parameter on the left to match the correct description on the right.

Note: One of the description on the right is a distractor and has no matching value.

RP/0/RP0/CPU0:router(config-bgp-a)# **bgp dampening 60 600 2400 240**

60	The penalty for each flap
600	Suppress a route when its penalty exceeds this value
2400	The amount of time for the penalty to decrease to one-half of its current value
240	If a flapping route penalty decreases and falls below this value, the route is unsuppressed.
	The maximum time a route can be suppressed

Answer:

Referring to the **bgp dampening** command shown below.

Drag the BGP route dampening configuration parameter on the left to match the correct description on the right.

Note: One of the description on the right is a distractor and has no matching value.

RP/0/RP0/CPU0:router(config-bgp-af)# **bgp dampening 60 600 2400 240**

60

600

2400

240

The penalty for each flap

2400

60

600

240

Explanation:

The amount of time for the penalty to decrease to one-half of its current value - 60

Suppress a route when its penalty exceeds this value - 2400

If a flapping route penalty decreases and falls below this value , the route is unsuppressed - 600

The maximum time a route can be suppressed - 240

QUESTION NO: 73

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on each of the router icon to gain access to the console of each router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation. If a certain command or command option is not supported, please try to use a different command that is supported.

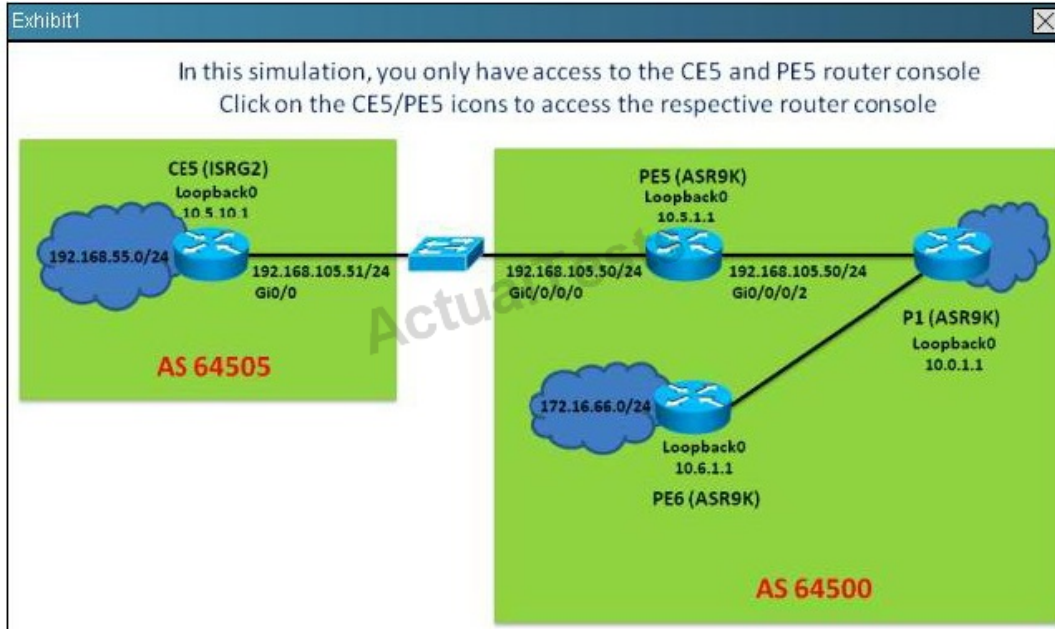
For example, the show running-config and the ping commands are **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE5 and PE5 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

Note: The CE5 router is an IOS router and the PE5 router is an IOS-XR router.





On the PE5 router, which statement is correct regarding the learned BGP prefixes?

- A. The 209.165.201.0/27 prefix is received from the 10.0.1.1 IBGP peer which is a route reflector
- B. The 172.16.66.0/24 prefix BGP next-hop points to the route reflector
- C. All prefixes learned on PE5 has the default local preference value
- D. The 209.165.202.128/27 prefix is originated by the 10.0.1.1 IBGP peer

Answer: C

Explanation: #show ip bgp-- check i tag for PE5

QUESTION NO: 74

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on each of the router icon to gain access to the console of each router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation. If a certain command or command option is not supported, please try to use a different command that is supported.

For example, the show running-config and the ping commands are **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

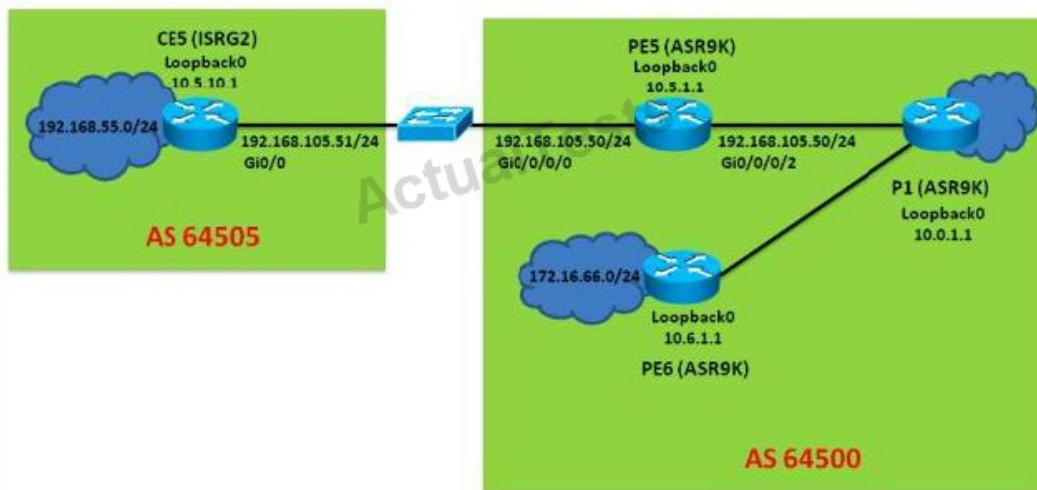
Scenario

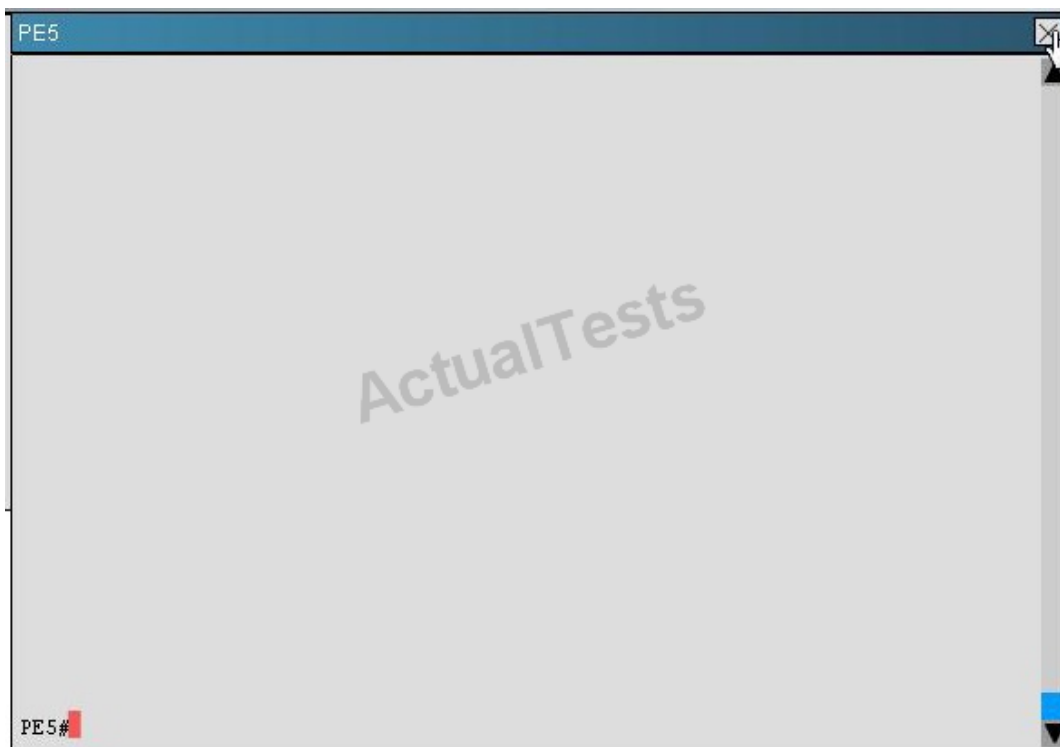
Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE5 and PE5 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

Note: The CE5 router is an IOS router and the PE5 router is an IOS-XR router.

Exhibit1

In this simulation, you only have access to the CE5 and PE5 router console
Click on the CE5/PE5 icons to access the respective router console





Which two statements regarding the BGP peerings are correct? (Choose two)

- A.** On PE5, the incoming prefixes received from the 192.168.105.51 EBGP peer is limited to a maximum of 10 prefixes
- B.** On PE5, the "rpln" inbound route policy is applied to the 192.168.105.51 EBGP peer
- C.** On PE5, the "pass" outbound route policy is applied to the 192.168.105.51 EBGP peer
- D.** PE5 has one EBGP peer (CE5) and two IBGP peers (P1 and PE6)

E. PE5 has received a total of 60 prefixes from its neighbors

Answer: A,E

Explanation: #show ip bgp

QUESTION NO: 75

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on each of the router icon to gain access to the console of each router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation. If a certain command or command option is not supported, please try to use a different command that is supported.

For example, the show running-config and the ping commands are **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

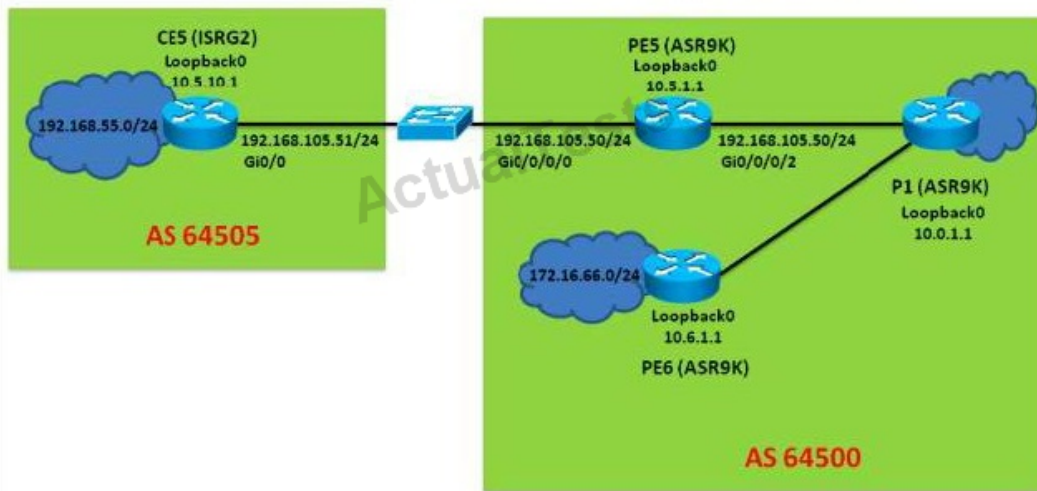
Scenario

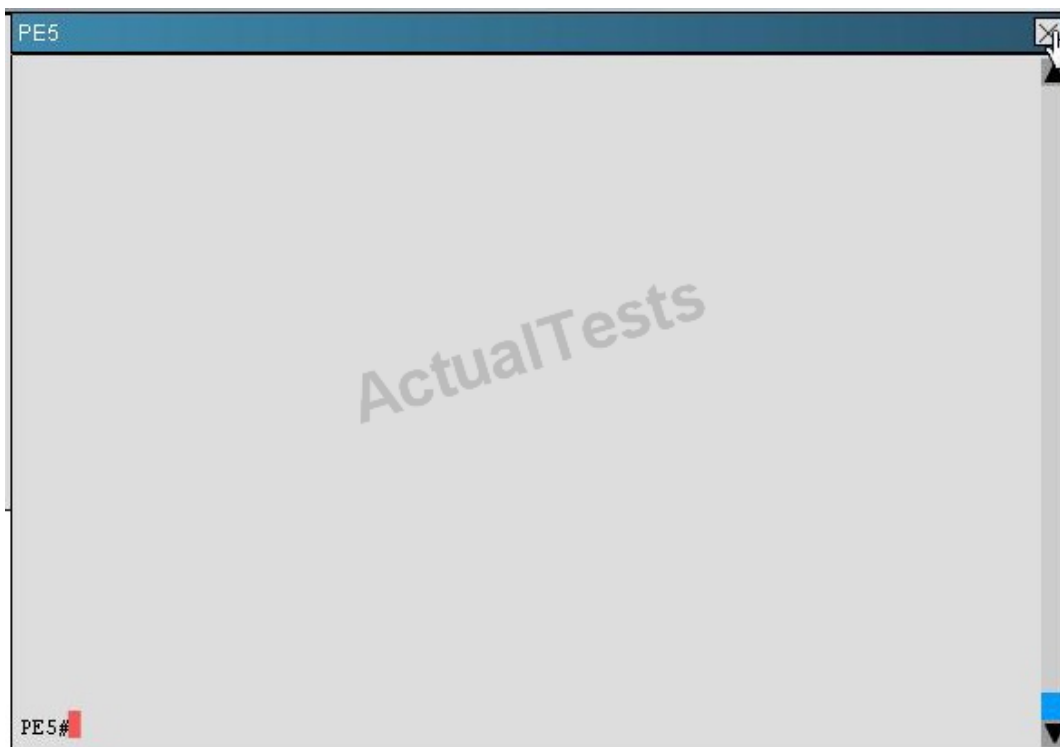
Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE5 and PE5 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

Note: The CE5 router is an IOS router and the PE5 router is an IOS-XR router.

Exhibit1

In this simulation, you only have access to the CE5 and PE5 router console
Click on the CE5/PE5 icons to access the respective router console





Which three statements regarding the BGP operations are correct? (Choose three)

- A. PE5 will set the local preferences 200 on all the prefixes sent to CE5
- B. PE5 will set the local preference to 200 on all the prefixes learned from CE5
- C. CE5 has received 5 prefixes from the PE5 EBGP peer
- D. CE5 has the BGP scan interval set to 30 seconds

- E. CE5 is announcing the 192.168.55.0/24 prefix via EBGp to the PE5 EBGp peer
 F. The AS-Path to reach the 209.165.202.128/27 prefix from CE5 is: 64500 64497 64498

Answer: C,E,F

Explanation: #sh ip bgp | be Network

#sh ip bgp

#show ip bgp neighbors

QUESTION NO: 76

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on each of the router icon to gain access to the console of each router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation. If a certain command or command option is not supported, please try to use a different command that is supported.

For example, the show running-config and the ping commands are **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

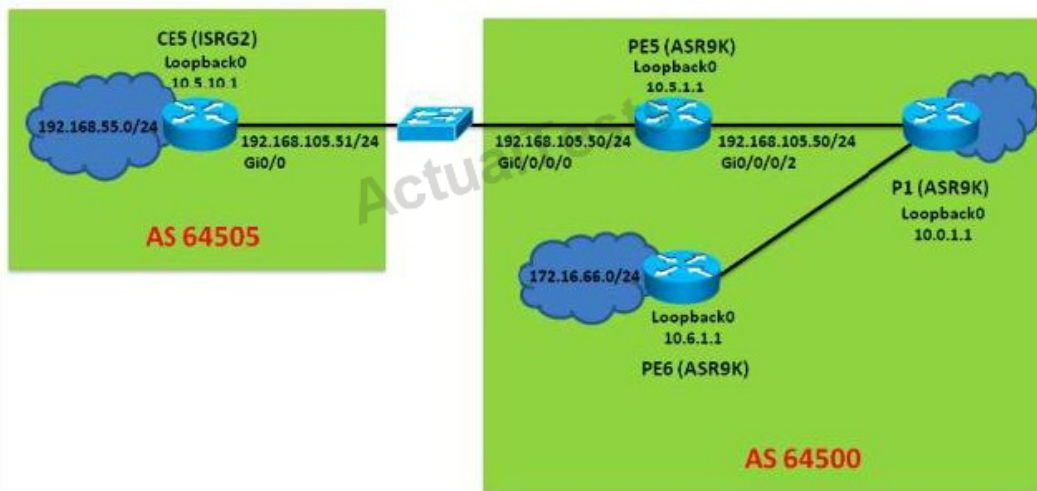
Scenario

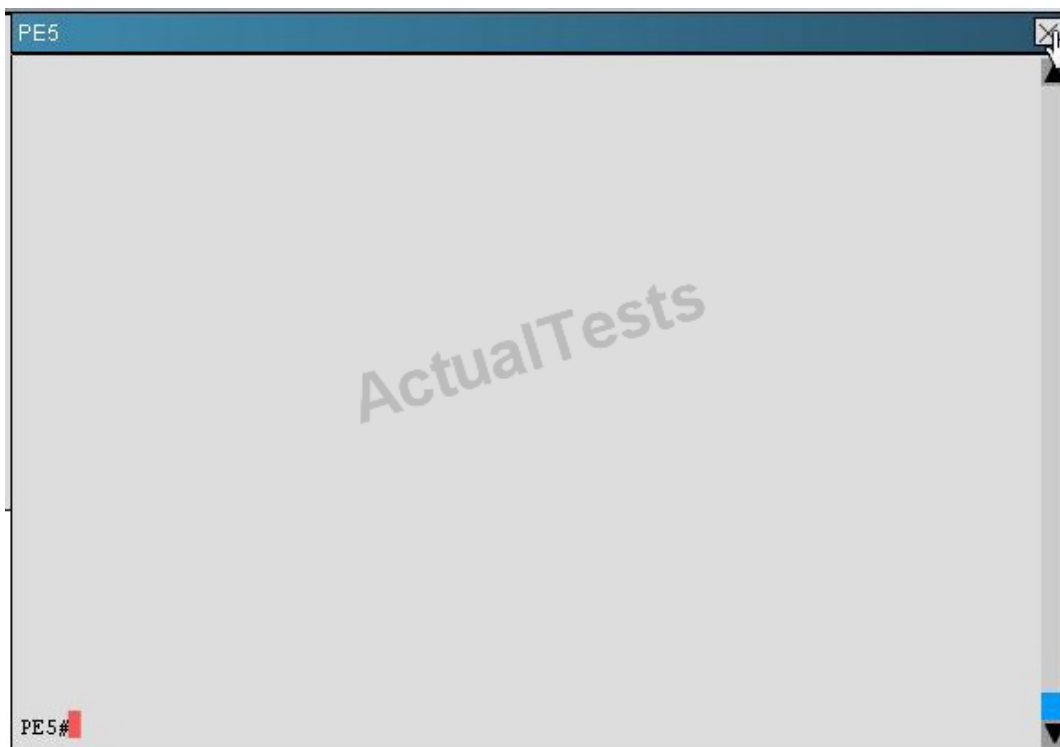
Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE5 and PE5 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

Note: The CE5 router is an IOS router and the PE5 router is an IOS-XR router.

Exhibit1

In this simulation, you only have access to the CE5 and PE5 router console
 Click on the CE5/PE5 icons to access the respective router console





Which three statements regarding the BGP operations are correct? (Choose three)

- A.** PE5 is the route reflector with P1 and PE6 as its client
- B.** PE5 is using the IS-IS route to reach the BGP next-hop for the 172.16.66.0/24 prefix
- C.** PE5 has BGP route dampening enabled
- D.** The BGP session between PE5 and P1 is established using the loopback interface and next-hop-self

E. The BGP session between PE5 and CE5 is established using the loopback interface

Answer: A,C,D

Explanation:

QUESTION NO: 77

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on each of the router icon to gain access to the console of each router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

For example, the show running-config and the ping commands are **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

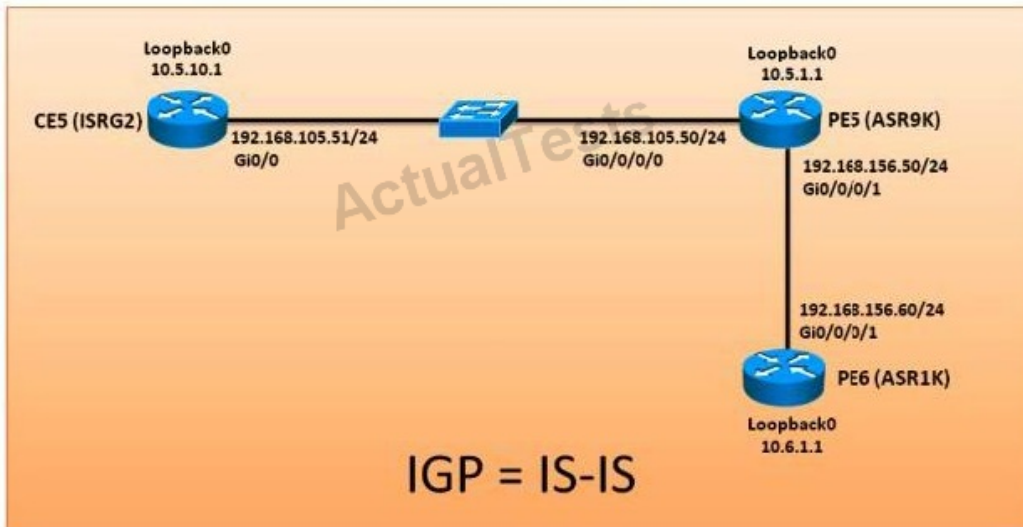
Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE5, PE5 and PE6 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

Note: The CE5 router is an IOS router, the PE5 router is an IOS-XR router, and the PE6 router is an IOS-XE router.

Exhibit1

Click on the CE5 and PE5 icons to access the respective router console
This simulation does not require access to the PE6 router



CE5

CE5#



Which router is configured as the RP for the 234.1.1.1 multicast group and which is the multicast source that is currently sending traffic to the 234.1.1.1 multicast group? (Choose two.)

- A. CE5
- B. PE5
- C. PE6
- D. 10.5.10.1
- E. 10.5.1.1
- F. 192.168.156.60

Answer: C,E

Explanation:

#show ip mroute 234.1.1.1

#show ip route

QUESTION NO: 78

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on each of the router icon to gain access to the console of each router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

For example, the show running-config and the ping commands are **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

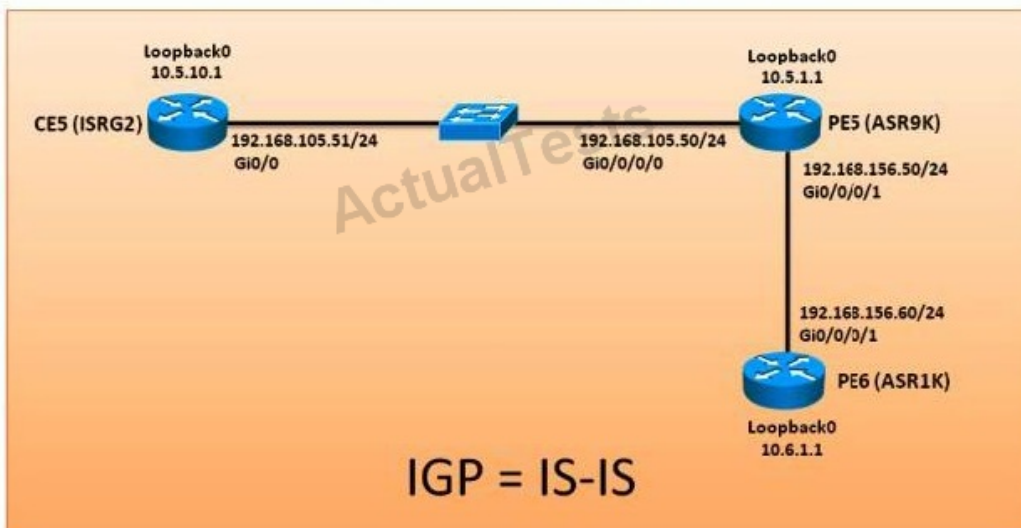
Scenario

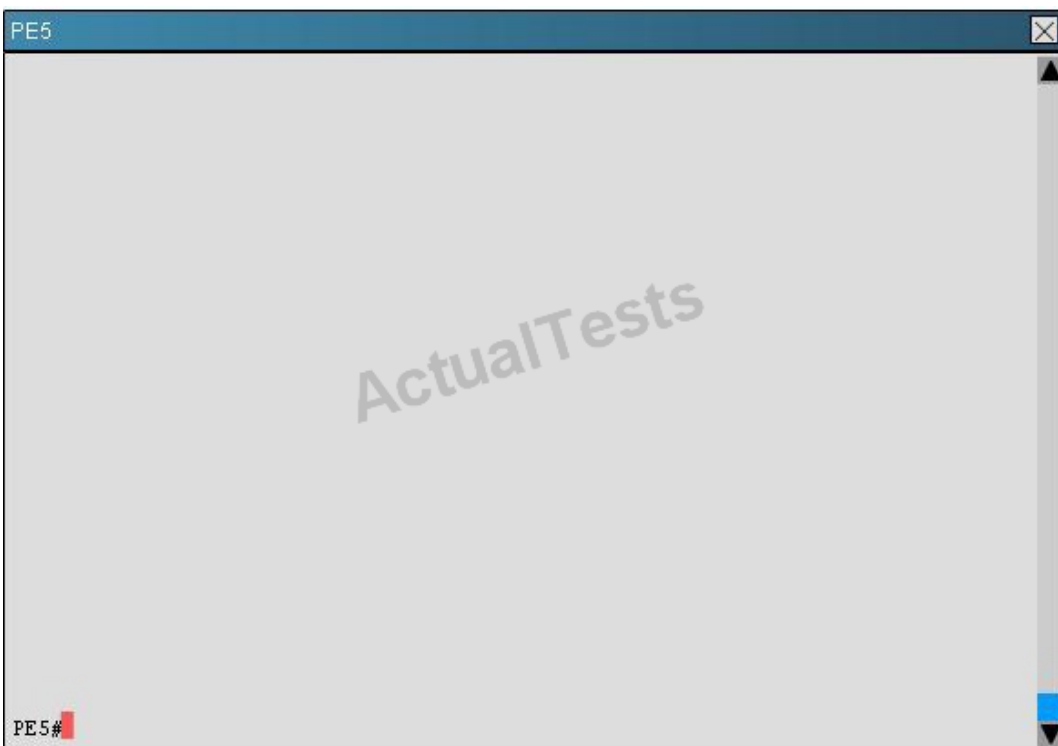
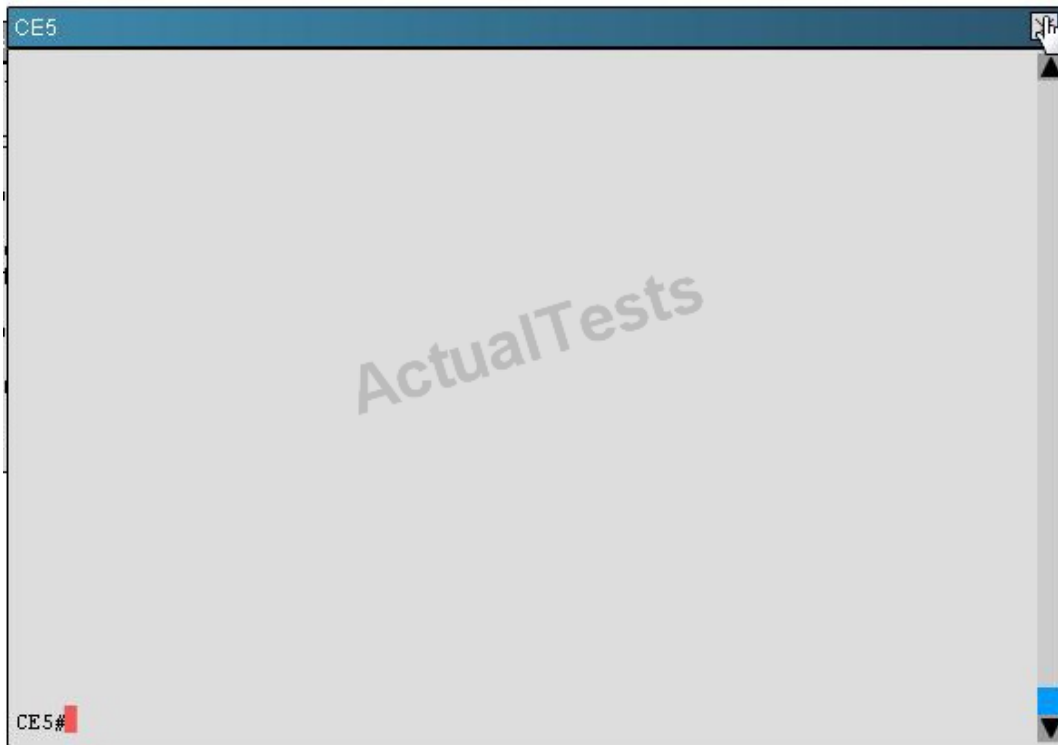
Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE5, PE5 and PE6 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

Note: The CE5 router is an IOS router, the PE5 router is an IOS-XR router, and the PE6 router is an IOS-XE router.

Exhibit1

Click on the CE5 and PE5 icons to access the respective router console
This simulation does not require access to the PE6 router





Which three statements are correct regarding the various multicast groups? (Choose three.)

- A. Currently there is no source sending traffic to the 224.1.1.1 multicast group
- B. PE5 has a Null OIL for the (*,224.0.1.40) entry
- C. PE5 has a Null OIL for the (*,224.1.1.1) entry
- D. CE5 has joined the 224.0.1.40 multicast group
- E. CE5 has a Null OIL for the (*,224.1.1.1) entry

Answer: C,D,E

Explanation: #show ip mroute

QUESTION NO: 79

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on each of the router icon to gain access to the console of each router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

For example, the show running-config and the ping commands are **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

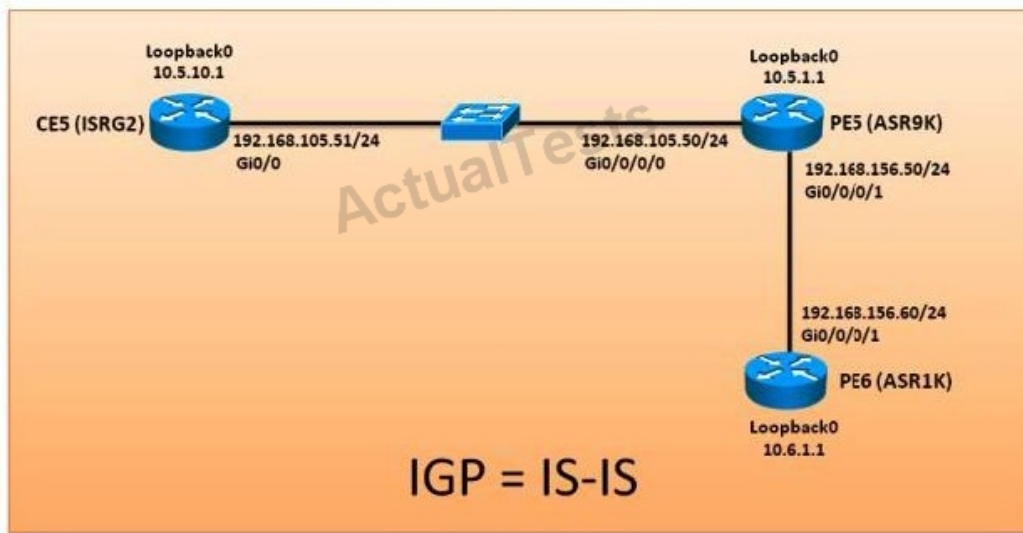
Scenario

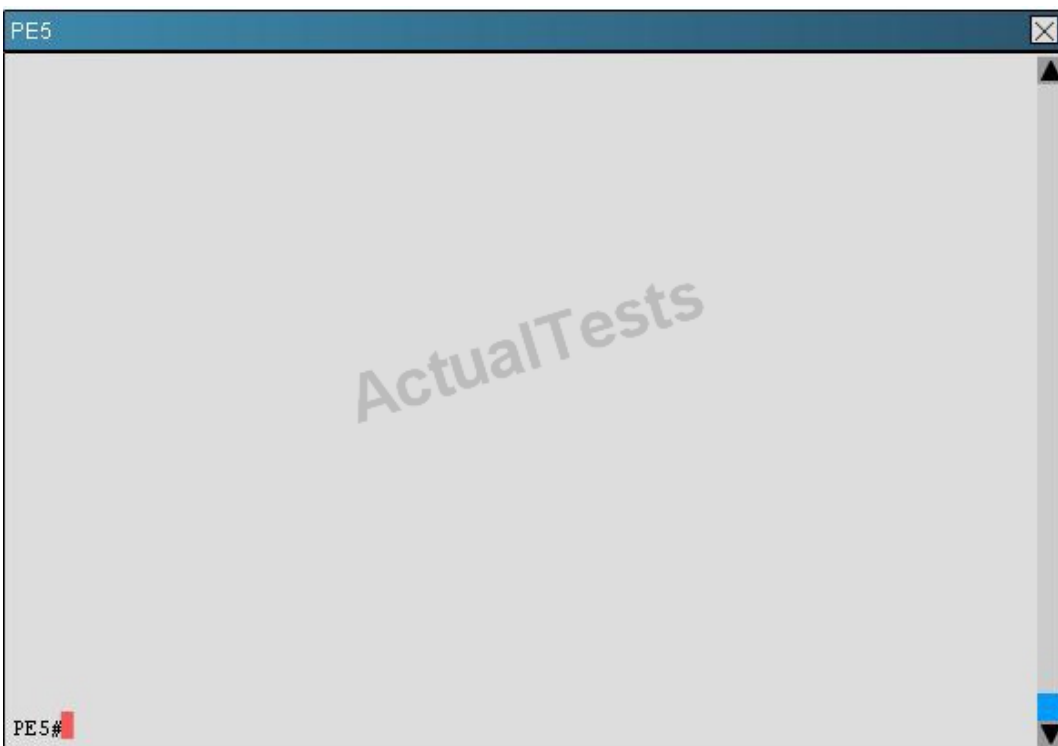
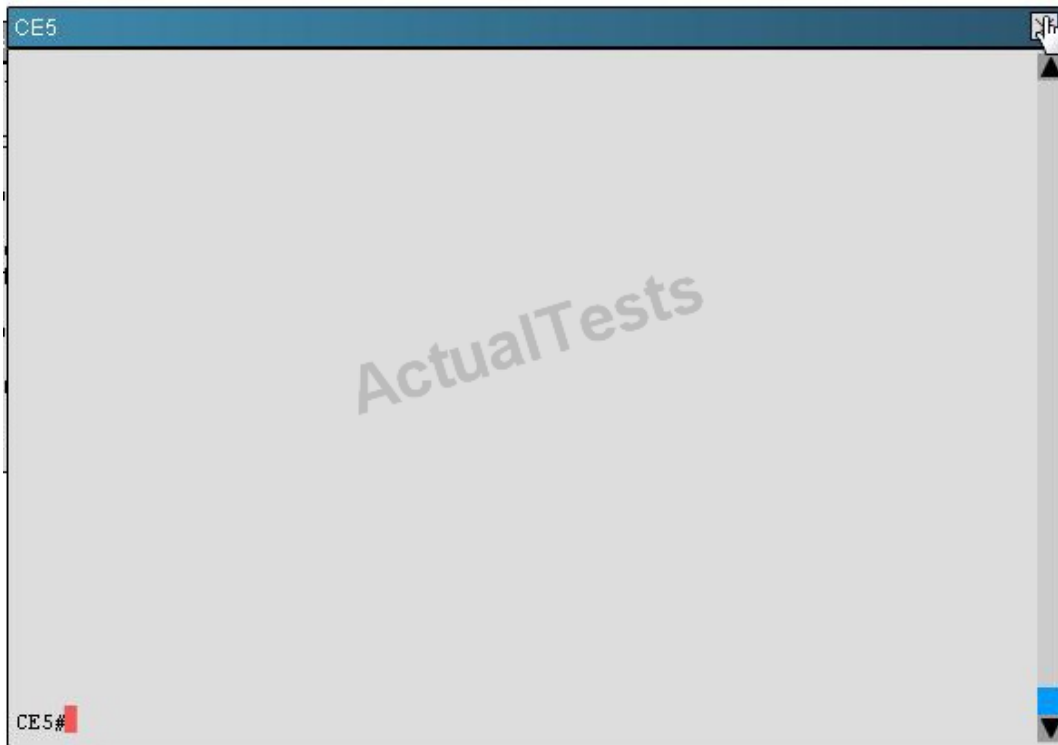
Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE5, PE5 and PE6 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

Note: The CE5 router is an IOS router, the PE5 router is an IOS-XR router, and the PE6 router is an IOS-XE router.

Exhibit1

Click on the CE5 and PE5 icons to access the respective router console
This simulation does not require access to the PE6 router





On the PE, which two statements are correct regarding the(192.168.156.60,224.1.1.1) entry?
(Choose two,)

- A.** The RPF neighbor points towards the RP
- B.** The RPF neighbor is reachable over the Gi0/0/0/1 interface
- C.** The OIL contains the Gi0/0/0/0 interface
- D.** The IIL is Null

Answer: A,C

Explanation: #show ip mroute

QUESTION NO: 80

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on each of the router icon to gain access to the console of each router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

For example, the show running-config and the ping commands are **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

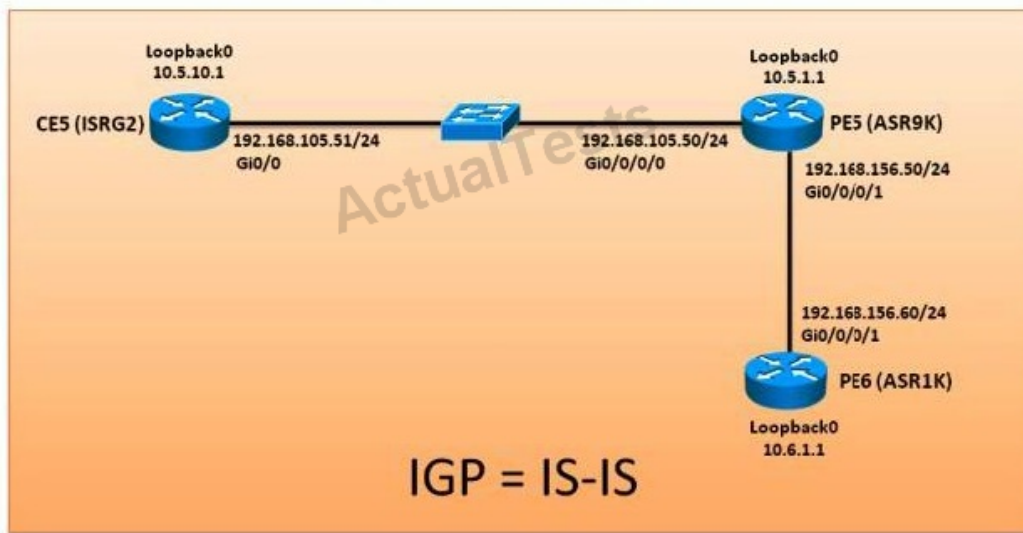
Scenario

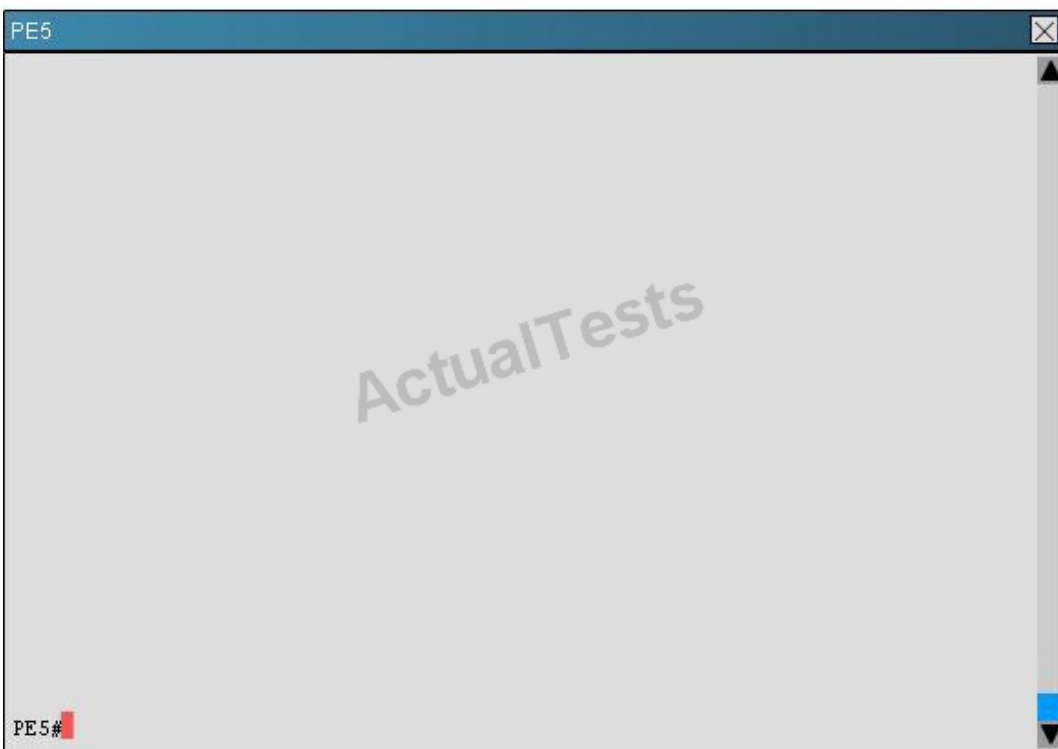
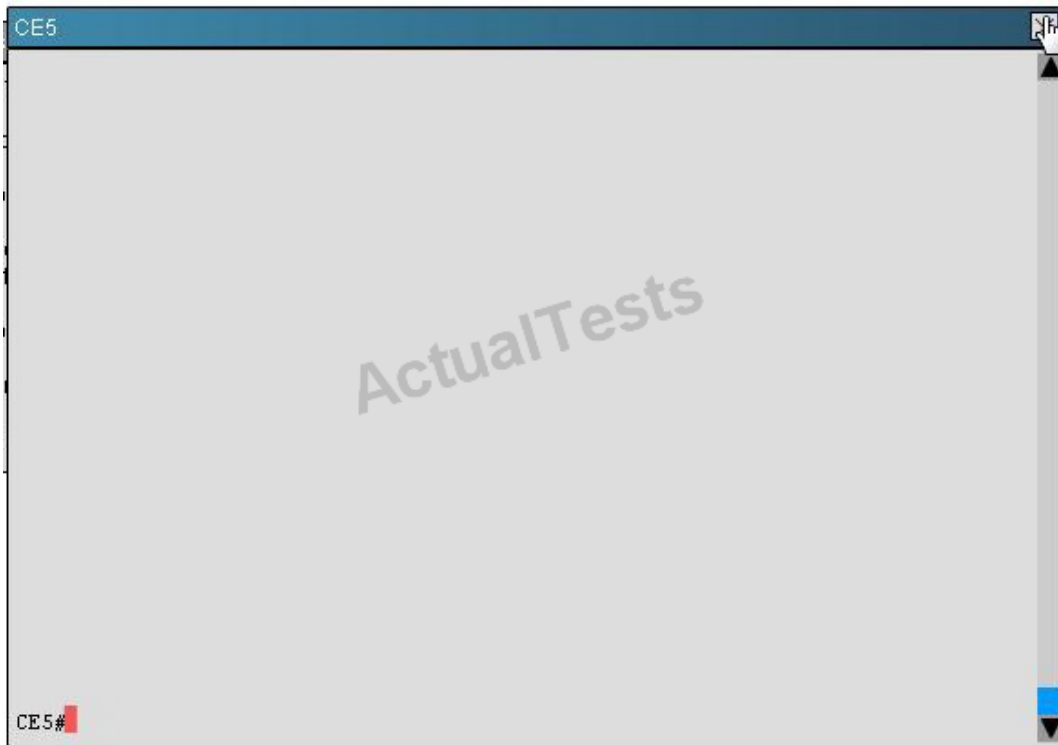
Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE5, PE5 and PE6 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

Note: The CE5 router is an IOS router, the PE5 router is an IOS-XR router, and the PE6 router is an IOS-XE router.

Exhibit1

Click on the CE5 and PE5 icons to access the respective router console
This simulation does not require access to the PE6 router





Which two statements are correct regarding the multicast operations on the router that is the RP?
(Choose two.)

- A. It is using IGMPv3
- B. The IGMP query interval is set to 125 seconds
- C. It is using the IPv4 unicast routing table to perform the RPF checks
- D. Static multicast routes are configured on the RP

Answer: A,C,D

Explanation: #show ip mroute

#show ip pim interface

#show ip igmp group

#show ip pim neighbor